

- 1 (i) Given that $y = \frac{\ln x}{x}$, find $\frac{dy}{dx}$ in terms of x . [2]

- (ii) Hence, or otherwise, find the exact value of $\int_1^e \frac{\ln x}{x^2} dx$, showing your working. [4]



<http://go.vjc.sg/2018p1q1>

Qn	Solution	Comments
1i	$y = \frac{\ln x}{x}$ $\frac{dy}{dx} = \frac{x\left(\frac{1}{x}\right) - \ln x}{x^2}$ $= \frac{1 - \ln x}{x^2}$	
1ii	Hence method, $\int_1^e \frac{\ln x}{x^2} dx = \int_1^e \frac{1}{x^2} - \frac{1 - \ln x}{x^2} dx$ $= \left[-\frac{1}{x} - \frac{\ln x}{x} \right]_1^e = -\left[\frac{1}{x} + \frac{\ln x}{x} \right]_1^e$ $= -\left(\frac{1}{e} + \frac{1}{e} - 0 - 1 \right)$ $= 1 - \frac{2}{e}$ Otherwise method, $\int_1^e \frac{\ln x}{x^2} dx = \left[-\frac{\ln x}{x} \right]_1^e + \int_1^e \frac{1}{x^2} dx$ $= -\frac{1}{e} - \left[\frac{1}{x} \right]_1^e$ $= -\frac{1}{e} - \frac{1}{e} + 1$ $= 1 - \frac{2}{e}$	Re-write $\frac{\ln x}{x^2}$ into $\frac{1}{x^2} - \frac{1 - \ln x}{x^2}$ to make use of the first part. The otherwise method is integration by parts. Do read the question to see if they allow “otherwise” methods. Usually, “otherwise” methods are harder and less obvious. Read the question. Keywords like “exact” and “showing your working” means you have to show clearly the integration. However, we can use GC to check that the answer is 0.26424.

2 Do not use a calculator in answering this question.

A curve has equation $y = \frac{3}{x}$ and a line has equation $y + 2x = 7$. The curve and the line intersect at the points A and B .

- (i) Find the x -coordinates of A and B . [2]
 (ii) Find the exact volume generated when the area bounded by the curve and the line is rotated about the x -axis through 360° . [4]



<http://go.vjc.sg/2018p1q2>

Qn	Solution	Comments
2	Sub $y = \frac{3}{x}$ into $y = -2x + 7$ $\frac{3}{x} = -2x + 7$ $3 = -2x^2 + 7x$ $2x^2 - 7x + 3 = 0$ $(2x - 1)(x - 3) = 0$ $x = \frac{1}{2} \text{ or } 3$ The x-coordinates of A and B are 3 and $\frac{1}{2}$.	Read the question. “Do not use a calculator...” However, we can use the calculator to check your answer. Good habit to check our answer by substituting into the first equation, i.e. $\frac{3}{x} = -2x + 7$.
ii		Good to draw a sketch to identify the region to be rotated – see that there is a hollow portion and we need to subtract = ans The volume generated is a difference of two volumes, so the working is

Qn	Solution	Comments
	$\text{Volume generated} = \pi \int_{\frac{1}{2}}^3 (-2x+7)^2 dx - \pi \int_{\frac{1}{2}}^3 \left(\frac{3}{x}\right)^2 dx$ $= \pi \left[\frac{(-2x+7)^3}{3(-2)} \right]_{\frac{1}{2}}^3 - \pi \left[-\frac{9}{x} \right]_{\frac{1}{2}}^3$ $= \pi \left(-\frac{1}{6} - \frac{6^3}{-6} \right) - \pi(-3+18)$ $= \pi \left(\frac{215}{6} - 15 \right)$ $= \frac{125}{6} \pi$	$\pi \int_a^b y_1^2 dx - \pi \int_a^b y_2^2 dx$, not $\pi \int_a^b (y_1 - y_2)^2 dx$. We can also use a calculator to check the result of your integration.

- 3 (i) It is given that $x \frac{dy}{dx} = 2y - 6$. Using the substitution $y = ux^2$, show that the differential equation can be transformed to $\frac{du}{dx} = f(x)$, where the function $f(x)$ is to be found. [3]
- (ii) Hence, given that $y = 2$ when $x = 1$, solve the differential equation $x \frac{dy}{dx} = 2y - 6$, to find y in terms of x . [4]



<http://go.vjc.sg/2018p1q3>

Qn	Solution	Comments
3i	$y = ux^2$ $\frac{dy}{dx} = \frac{du}{dx} x^2 + 2xu$ $x \frac{dy}{dx} = 2y - 6$ $x \left(\frac{du}{dx} x^2 + 2xu \right) = 2ux^2 - 6$ $x^3 \frac{du}{dx} + 2x^2 u = 2ux^2 - 6$ $\frac{du}{dx} = -\frac{6}{x^3} \text{ (shown)}$	Need to remember that u is not a constant!

Qn	Solution	Comments
ii	$\frac{du}{dx} = -\frac{6}{x^3}$ $\int du = \int -\frac{6}{x^3} dx$ $u = \frac{3}{x^2} + C$ $\frac{y}{x^2} = \frac{3}{x^2} + C$ $y = Cx^2 + 3$ When $x = 1, y = 2$ $2 = C + 3 \Rightarrow C = -1$ $\therefore y = -x^2 + 3$	“Hence” so we solve this $\frac{du}{dx} = -\frac{6}{x^3}$ first. Please remember the arbitrary constant when solving indefinite integrals. Read the question, we want the answer as “y in terms of x”, so we need to make y the subject.

4 (i) Find the exact roots of the equation $|2x^2 + 3x - 2| = 2 - x$. [4]

(ii) On the same axes, sketch the curves with equations $y = |2x^2 + 3x - 2|$ and $y = 2 - x$.

Hence solve exactly the inequality

$$|2x^2 + 3x - 2| < 2 - x. \quad [4]$$

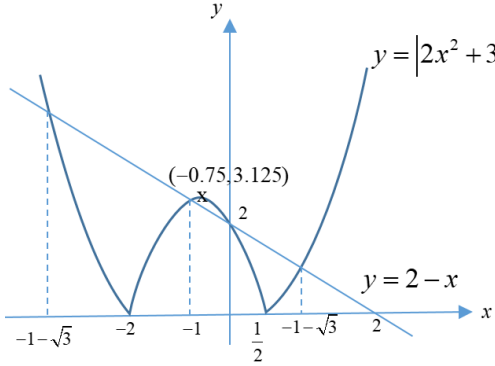


<http://go.vjc.sg/2018p1q4i>



<http://go.vjc.sg/2018p1q4ii>

Qn	Solution	Comments
4i	$ 2x^2 + 3x - 2 = 2 - x$ $2x^2 + 3x - 2 = 2 - x \text{ or } 2x^2 + 3x - 2 = -(2 - x)$ Consider	Once again, notice the word “exact – so we can’t use GC/graph BUT we can “use” the graph in (ii) to check our answer. Recall the idea that if $x = k$ then $x = k$ or $x = -k$. Apply this

Qn	Solution	Comments
	$2x^2 + 3x - 2 = 2 - x$ $2x^2 + 4x - 4 = 0$ $x^2 + 2x - 2 = 0$ $x = \frac{-2 \pm \sqrt{4+8}}{2}$ $= -1 \pm \sqrt{3}$ and then consider, $2x^2 + 3x - 2 = -(2 - x)$ $2x^2 + 2x = 0$ $x(x+1) = 0$ $x = 0 \text{ or } -1$ Sub $x = 0, -1, -1 + \sqrt{3}, -1 - \sqrt{3}$ into the equation $2x^2 + 3x - 2 = 2 - x$ to check if they satisfy the original equation. Since all $x = 0, -1, -1 + \sqrt{3}, -1 - \sqrt{3}$ satisfy the equation $2x^2 + 3x - 2 = 2 - x$, all solutions are accepted.	concept to solve the equations. The last step to verify that all the solutions satisfy the equation is necessary as the method we employ might lead us to solutions that are not applicable (e.g. try $x = x^2 - 1$).
ii	 From graph, when $2x^2 + 3x - 2 < 2 - x$, $-1 - \sqrt{3} < x < -1 \text{ or } 0 < x < -1 + \sqrt{3}.$	Our graph in this part should show 4 points of intersection, corresponding to the 4 answers we found in (i).

5 Functions f and g are defined by

$$f : x \mapsto \frac{x+a}{x+b} \text{ for } x \in \mathbb{R}, \quad x \neq -b, \quad a \neq -1,$$

$$g : x \mapsto x \text{ for } x \in \mathbb{R}.$$

It is given that $ff = g$.

Find the value of b .

Find $f^{-1}(x)$ in terms of x and a .

[5]



<http://go.vjc.sg/2018p1q5>

Qn	Solution	Comments
5	$ff(x) = \frac{\frac{x+a}{x+b} + a}{\frac{x+a}{x+b} + b}$ $= \frac{x+a+a(x+b)}{x+a+b(x+b)}$ $= \frac{x+ax+a+ab}{x+bx+a+b^2}$ Since $ff(x) = x$, $\frac{x+ax+a+ab}{x+bx+a+b^2} = x$ $x+ax+a+ab = x(x+bx+a+b^2)$ $= x^2 + bx^2 + ax + b^2x$ $(1+b)x^2 + (b^2-1)x = a(1+b)$ Therefore, the coefficients of x^2 and x needs to be 0. $\therefore b = -1$ Let $y = \frac{x+a}{x-1}$ $y(x-1) = x+a$ $xy - y = x+a$ $xy - x = y+a$ $x = \frac{y+a}{y-1}$ $\therefore f^{-1}(x) = \frac{x+a}{x-1}$ Alternatively, Since $ff(x) = x$ $f^{-1}[ff(x)] = f^{-1}(x)$ $f(x) = f^{-1}(x)$ $\therefore f^{-1}(x) = \frac{x+a}{x-1}$	By right, the question should be $ff(x) = g(x)$. Since the domains of ff and g are different, the 2 functions cannot be equal. Our goal here is to adjust the values of a and b such that $\frac{x+ax+a+ab}{x+bx+a+b^2}$ becomes x. FYI: The condition that $a \neq -1$ is to make sure that $f(x) \neq 1$. There is no need to find domain of f^{-1} for this question. A shortcut: Since we know that f^{-1} exists, and $ff(x) = x$, we can take f^{-1} on both sides and get $f(x) = f^{-1}(x)$.

6 Vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are such that $\mathbf{a} \neq \mathbf{0}$ and $\mathbf{a} \times 3\mathbf{b} = 2\mathbf{a} \times \mathbf{c}$.

(i) Show that $3\mathbf{b} - 2\mathbf{c} = \lambda\mathbf{a}$, where λ is a constant. [2]

(ii) It is now given that $\mathbf{a}$ and $\mathbf{c}$ are unit vectors, that the modulus of $\mathbf{b}$ is 4 and that the angle between $\mathbf{b}$ and $\mathbf{c}$ is 60° . Using a suitable scalar product, find exactly the two possible values of λ . [5]



<http://go.vjc.sg/2018p1q6>

Qn	Solution	Comments
6i	$\mathbf{a} \times 3\mathbf{b} = 2\mathbf{a} \times \mathbf{c}$ $\mathbf{a} \times 3\mathbf{b} - \mathbf{a} \times 2\mathbf{c} = \mathbf{0}$ $\mathbf{a} \times (3\mathbf{b} - 2\mathbf{c}) = \mathbf{0}$ Since $\mathbf{a} \neq \mathbf{0}$, $\mathbf{a}$ is parallel to $3\mathbf{b} - 2\mathbf{c}$ or $3\mathbf{b} - 2\mathbf{c} = \mathbf{0}$ $\therefore 3\mathbf{b} - 2\mathbf{c} = \lambda\mathbf{a}$, where λ is a constant. (shown)	Recall the conditions for the cross product of 2 vectors to be $\mathbf{0}$.
ii	$3\mathbf{b} - 2\mathbf{c} = \lambda\mathbf{a}$ $(3\mathbf{b} - 2\mathbf{c}) \cdot (3\mathbf{b} - 2\mathbf{c}) = (\lambda\mathbf{a}) \cdot (\lambda\mathbf{a})$ $9 \mathbf{b} ^2 - 12\mathbf{b} \cdot \mathbf{c} + 4 \mathbf{c} ^2 = \lambda^2 \mathbf{a} ^2$ $9(16) - 12 \mathbf{b} \mathbf{c} \cos 60^\circ + 4 = \lambda^2$ $9(16) - 12(4)(1)(0.5) + 4 = \lambda^2$ $\lambda^2 = 124$ $\lambda = \pm 2\sqrt{31}$	Since we know the angle between $\mathbf{b}$ and $\mathbf{c}$, we can find $\mathbf{b} \cdot \mathbf{c}$. We now have moduli of $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{b} \cdot \mathbf{c}$. The motivation for this question becomes how to express $3\mathbf{b} - 2\mathbf{c} = \lambda\mathbf{a}$ (since we have to find the value of λ) in terms of the 4 values. One strategy would be to do self dot product, i.e. $(3\mathbf{b} - 2\mathbf{c}) \cdot (3\mathbf{b} - 2\mathbf{c}) = (\lambda\mathbf{a}) \cdot (\lambda\mathbf{a})$ Since, this ends up with an expression containing values such as $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{b} \cdot \mathbf{c}$.

7 A curve C has equation $\frac{x^2 - 4y^2}{x^2 + xy^2} = \frac{1}{2}$.

(i) Show that $\frac{dy}{dx} = \frac{2x - y^2}{2xy + 16y}$. [3]

The points P and Q on C each have x -coordinate 1. The tangents to C at P and Q meet at the point N .

(ii) Find the exact coordinates of N . [6]



<http://go.vjc.sg/2018p1q7>

Qn	Solution	Comments
7i	$\frac{x^2 - 4y^2}{x^2 + xy^2} = \frac{1}{2}$ $2x^2 - 8y^2 = x^2 + xy^2$ $x^2 - 8y^2 = xy^2$ $2x - 16y \frac{dy}{dx} = y^2 + 2xy \frac{dy}{dx}$ $(2xy + 16y) \frac{dy}{dx} = 2x - y^2$ $\frac{dy}{dx} = \frac{2x - y^2}{2xy + 16y} \text{ (shown)}$	You could also attempt this question by differentiating both sides w.r.t. x directly using quotient rule.
ii	When $x = 1$, $\frac{1 - 4y^2}{1 + y^2} = \frac{1}{2}$ $2 - 8y^2 = 1 + y^2$ $9y^2 = 1$ $y = \pm \frac{1}{3}$ The coordinates of P and Q are $\left(1, \frac{1}{3}\right)$ and $\left(1, -\frac{1}{3}\right)$. At $\left(1, \frac{1}{3}\right)$, $\frac{dy}{dx} = \frac{2x - y^2}{2xy + 16y} = \frac{2 - \frac{1}{9}}{\frac{2}{3} + \frac{16}{3}} = \frac{17}{54}$	x-coordinate = 1, we find the y-coordinate of P and Q first. Next is to work out the equations of the tangent line to C at P and Q. Solving the equations of the tangent lines simultaneously will give the exact coordinates of N.

Qn	Solution	Comments
	$y - \frac{1}{3} = \frac{17}{54}(x-1)$ $y = \frac{17}{54}x + \frac{1}{54} \text{ ----- (1)}$ At $\left(1, -\frac{1}{3}\right)$, $\frac{dy}{dx} = \frac{2x - y^2}{2xy + 16y} = \frac{2 - \frac{1}{9}}{-\frac{2}{3} - \frac{16}{3}} = -\frac{17}{54}$ $y + \frac{1}{3} = -\frac{17}{54}(x-1)$ $y = -\frac{17}{54}x - \frac{1}{54} \text{ ----- (2)}$ Solving equations (1) and (2) $\frac{17}{54}x + \frac{1}{54} = -\frac{17}{54}x - \frac{1}{54}$ $\frac{17}{27}x = -\frac{1}{27}$ $x = -\frac{1}{17}$ $y = -\frac{17}{54}\left(-\frac{1}{17}\right) - \frac{1}{54} = 0$ The exact coordinates of N are $\left(-\frac{1}{17}, 0\right)$.	

- 8 A sequence $u_1, u_2, u_3, \dots$ is such that $u_{n+1} = 2u_n + An$, where A is a constant and $n \geq 1$.

(i) Given that $u_1 = 5$ and $u_2 = 15$, find A and u_3 . [2]

It is known that the n th term of this sequence is given by

$$u_n = a(2^n) + bn + c.$$

where a , b and c are constants.

(ii) Find a , b and c . [4]

(iii) Find $\sum_{r=1}^n u_r$ in terms of n . (You need not simplify your answer.) [4]



<http://go.vjc.sg/2018p1q8>

Qn	Solution	Comments
8i	Sub $n = 1$ into $u_{n+1} = 2u_n + An$ $u_2 = 2u_1 + A(1)$ $15 = 2(5) + A$ $A = 5$ When $n = 2$, $u_3 = 2u_2 + 5(2)$ $= 2(15) + 5(2)$ $= 40$	
ii	Substituting $n = 1, 2$ and 3 into $u_n = a(2^n) + bn + c$: $2a + b + c = 5$ $4a + 2b + c = 15$ $8a + 3b + c = 40$ By GC, $a = 7.5, b = -5, c = -5$	
iii	$u_n = 7.5(2^n) - 5n - 5$ $\sum_{r=1}^n (7.5(2^r) - 5r - 5) = 7.5 \sum_{r=1}^n 2^r - 5 \sum_{r=1}^n r - \sum_{r=1}^n 5$ $= 7.5 \left(\frac{2(2^n - 1)}{2 - 1} \right) - 5 \left(\frac{n(1 + n)}{2} \right) - 5n$ $= 15(2^n - 1) - 5n \left(\frac{1}{2}(1 + n) + 1 \right)$ $= 15(2^n - 1) - \frac{5n}{2}(3 + n)$	$\sum_{r=1}^n 2^r$ and $\sum_{r=1}^n r$ are GP and AP respectively, so we apply the sum formula for these two series accordingly. While the question say you need not simplify fully, leaving your answer as “ $2 - 1$ ” is not acceptable.

9 A curve C has parametric equations

$$x = 2\theta - \sin 2\theta$$

$$y = 2 \sin^2 \theta$$

for $0 \leq \theta \leq \pi$.

(i) Show that $\frac{dy}{dx} = \cot \theta$. [4]

(ii) The normal to the curve at the point where $\theta = \alpha$ meets the x -axis at the point A . Show that the x -coordinate of A is $k\alpha$, where k is a constant to be found. [4]

(iii) Do not use a calculator in answering this part.

The distance between two points along a curve is the arc-length. Scientists and engineers need to use arc-length in applications such as finding the work done in moving an object along the path described by a curve or the length of cabling used on a suspension bridge.

The arc-length between two points on C , where $\theta = \beta$ and $\theta = \gamma$, is given by the formula

$$\int_{\beta}^{\gamma} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta.$$

Find the total length of C .

[5]



<http://go.vjc.sg/2018p1q9>

Qn	Solution	Comments
9i	$x = 2\theta - \sin 2\theta \qquad y = 2\sin^2 \theta$ $\frac{dx}{d\theta} = 2 - 2\cos 2\theta \qquad \frac{dy}{d\theta} = 4\sin \theta \cos \theta$ $\frac{dy}{dx} = \frac{4\sin \theta \cos \theta}{2 - 2\cos 2\theta}$ $= \frac{4\sin \theta \cos \theta}{2 - 2(1 - 2\sin^2 \theta)}$ $= \frac{4\sin \theta \cos \theta}{4\sin^2 \theta}$ $= \cot \theta$	
ii	When $\theta = \alpha$, $x = 2\alpha - \sin 2\alpha$, $y = 2\sin^2 \alpha$, $\frac{dy}{dx} = \cot \alpha$ Gradient of normal $= -\frac{1}{\cot \alpha} = -\tan \alpha$ $y - 2\sin^2 \alpha = -\tan \alpha (x - 2\alpha + \sin 2\alpha)$ $= -x \tan \alpha + 2\alpha \tan \alpha - \frac{\sin \alpha}{\cos \alpha} (2\sin \alpha \cos \alpha)$ $= -x \tan \alpha + 2\alpha \tan \alpha - 2\sin^2 \alpha$ $y = -x \tan \alpha + 2\alpha \tan \alpha$ When $y = 0$, $-x \tan \alpha + 2\alpha \tan \alpha = 0$ $x = 2\alpha$ The x-coordinate of A is 2α.	

Qn	Solution	Comments
iii	$\begin{aligned} \text{Length of } C &= \int_0^\pi \sqrt{(4\sin^2 \theta)^2 + (4\sin \theta \cos \theta)^2} \, d\theta \\ &= \int_0^\pi \sqrt{16\sin^4 \theta + 16\sin^2 \theta \cos^2 \theta} \, d\theta \\ &= \int_0^\pi \sqrt{16\sin^2 \theta (\sin^2 \theta + \cos^2 \theta)} \, d\theta \\ &= \int_0^\pi \sqrt{16\sin^2 \theta} \, d\theta \\ &= \int_0^\pi 4\sin \theta \, d\theta \end{aligned}$ Since $4\sin \theta \geq 0$ for $0 \leq \theta \leq \pi$, $\begin{aligned} \text{Length of } C &= \int_0^\pi 4\sin \theta \, d\theta \\ &= [-4\cos \theta]_0^\pi \\ &= -4(-1) + 4(1) \\ &= 8 \end{aligned}$	This question is unfamiliar, so we will just follow instructions – sub in $\frac{dy}{d\theta}$, $\frac{dx}{d\theta}$, β and γ. Total length of C means we are using $0 \leq \theta \leq \pi$, as stated in the first part of the question. Recall that $\sqrt{x^2} = x$, and $x = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$ We need to check the range of values of θ in order to ascertain that $4\sin \theta = 4\sin \theta$.

- 10** An electrical circuit comprises a power source of V volts in series with a resistance of R ohms, a capacitance of C farads and an inductance of L henries. The current in the circuit, t seconds after turning on the power, is I amps and the charge on the capacitor is q coulombs. The circuit can be used by scientists to investigate resonance, to model heavily damped motion and to tune into radio stations on a stereo tuner. It is given that R , C and L are constants, and that $I = 0$ when $t = 0$.

A differential equation for the circuit is $L \frac{dI}{dt} + RI + \frac{q}{C} = V$, where $I = \frac{dq}{dt}$.

- (i) Show that, under certain conditions on V which should be stated,

$$L \frac{d^2 I}{dt^2} + R \frac{dI}{dt} + \frac{I}{C} = 0. \quad [2]$$

It is now given that the differential equation in part (i) holds for the rest of the question.

- (ii) Given that $I = Ate^{-\frac{Rt}{2L}}$ is a solution of the differential equation, where A is a positive constant, show that $C = \frac{4L}{R^2}$. [5]

- (iii) In a particular circuit, $R = 4$, $L = 3$ and $C = 0.75$. Find the maximum value of I in terms of A , showing that this value is a maximum. [4]

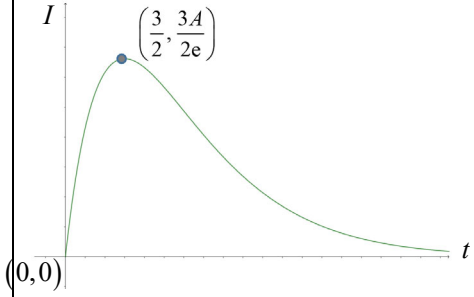
- (iv) Sketch the graph of I against t . [2]



<http://go.vjc.sg/2018p1q10>

Qn	Solution	Comments
10i	$L \frac{dI}{dt} + RI + \frac{q}{C} = V$ Differentiate w.r.t. t $L \frac{d^2 I}{dt^2} + R \frac{dI}{dt} + \frac{1}{C} \left(\frac{dq}{dt} \right) = \frac{dV}{dt}$ $L \frac{d^2 I}{dt^2} + R \frac{dI}{dt} + \frac{I}{C} = \frac{dV}{dt}$ If the power source is providing constant voltage, then V is a constant and $\frac{dV}{dt} = 0$. This leads to $L \frac{d^2 I}{dt^2} + R \frac{dI}{dt} + \frac{I}{C} = 0 \quad (\text{shown})$	This part has many letters, please be aware which are constants and which are variables!
ii	$I = Ate^{-\frac{Rt}{2L}}$ $\frac{dI}{dt} = Ae^{-\frac{Rt}{2L}} + At \left(-\frac{R}{2L} \right) e^{-\frac{Rt}{2L}}$ $= Ae^{-\frac{Rt}{2L}} \left(1 - \frac{Rt}{2L} \right)$ $\frac{d^2 I}{dt^2} = Ae^{-\frac{Rt}{2L}} \left(-\frac{R}{2L} \right) - \frac{R}{2L} Ae^{-\frac{Rt}{2L}} \left(1 - \frac{Rt}{2L} \right)$ $= Ae^{-\frac{Rt}{2L}} \left(-\frac{R}{L} + \left(\frac{R}{2L} \right)^2 t \right)$ Sub the derivatives into $L \frac{d^2 I}{dt^2} + R \frac{dI}{dt} + \frac{I}{C} = 0$ $L \left(Ae^{-\frac{Rt}{2L}} \left(-\frac{R}{L} + \left(\frac{R}{2L} \right)^2 t \right) \right) + R \left(Ae^{-\frac{Rt}{2L}} \left(1 - \frac{Rt}{2L} \right) \right) + \frac{Ate^{-\frac{Rt}{2L}}}{C} = 0$ Since $A > 0$ and $e^{-\frac{Rt}{2L}} > 0$,	We do not need to show that $I = Ate^{-\frac{Rt}{2L}}$. We are supposed to use the solution to find an expression for C. We need to check that $Ae^{-\frac{Rt}{2L}} \neq 0$ before dividing.

Qn	Solution	Comments
	$L\left(-\frac{R}{L} + \left(\frac{R}{2L}\right)^2 t\right) + R\left(1 - \frac{Rt}{2L}\right) + \frac{t}{C} = 0$ $-R + \frac{R^2}{4L}t + R - \frac{R^2t}{2L} + \frac{t}{C} = 0$ $\left(-\frac{R^2}{4L} + \frac{1}{C}\right)t = 0$ Since this equation holds for all $t \geq 0$, $-\frac{R^2}{4L} + \frac{1}{C} = 0$ $C = \frac{4L}{R^2}$	
iii	In a particular circuit, $R = 4$, $L = 3$ and $C = 0.75$. Note that $C = \frac{4(3)}{(4)^2} = 0.75$ Hence $I = Ate^{-\frac{2t}{3}}$ is a solution to the DE $I = Ate^{-\frac{2t}{3}}$ $\frac{dI}{dt} = Ae^{-\frac{2t}{3}} - \frac{2}{3}Ate^{-\frac{2t}{3}}$ When $\frac{dI}{dt} = 0$, $Ae^{-\frac{2t}{3}} - \frac{2}{3}Ate^{-\frac{2t}{3}} = 0$ $1 - \frac{2}{3}t = 0$ $t = \frac{3}{2}$ $\max I = \frac{3A}{2e}$ To show it is a maximum: From $L\frac{d^2I}{dt^2} + R\frac{dI}{dt} + \frac{I}{C} = 0$ Given that $R = 4$, $L = 3$ and $C = 0.75$ When $t = 1.5$, $\frac{dI}{dt} = 0$,	Since we know that $Ae^{-\frac{Rt}{2L}} \neq 0$, we can just divide.

Qn	Solution	Comments
	$3\frac{d^2I}{dt^2} + 0 + \frac{A(1.5)e^{\frac{1.5(4)}{2(3)}}}{0.75} = 0$ $\frac{d^2I}{dt^2} = -2Ae^{-1} \times \frac{1}{3}$ $= -\frac{2}{3}Ae^{-1}$ Since $A > 0$ and $e^{-1} > 0$, $\frac{d^2I}{dt^2} < 0$. Hence, the value of I is a maximum.	
iv		Our graph should pass through the origin (as stated in the question). It should also have the maximum point that we found in (iii). While we can use the GC to sketch the curve, we need to recall that t cannot take negative values. Note that we assume that $R = 4$, $L = 3$ and $C = 0.75$ still apply for this part

- 11** Mr Wong is considering investing money in a savings plan. One plan, P , allows him to invest \$100 into the account on the first day of every month. At the end of each month the total in the account is increased by $a\%$.

(i) It is given that $a = 0.2$.

(a) Mr Wong invests \$100 on 1 January 2016. Write down how much this \$100 is worth at the end of 31 December 2016. [1]

(b) Mr Wong invests \$100 on the first day of each of the 12 months of 2016. Find the total amount in the account at the end of 31 December 2016. [3]

(c) Mr Wong continues to invest \$100 on the first day of each month. Find the month in which the total in the account will first exceed \$3000. Explain whether this occurs on the first or last day of the month. [5]

An alternative plan, Q , also allows him to invest \$100 on the first day of every month. Each \$100 invested earns a fixed bonus of \$ b at the end of every month for which it has been in the account. This bonus is added to the account. The accumulated bonuses themselves do not earn any further bonus.

(ii) (a) Find, in terms of b , how much \$100 invested on 1 January 2016 will be worth at the end of 31 December 2016. [1]

- (b) Mr Wong invests \$100 on the first day of each of the 24 months in 2016 and 2017. Find the value of b such that the total value of all the investments, including bonuses, is worth \$2800 at the end of 31 December 2017. [3]

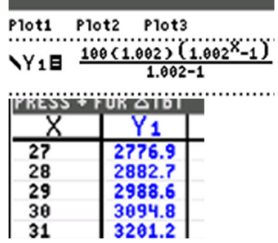
It is given instead that $a = 1$ for plan P .

- (iii) Find the value of b for plan Q such that both plans give the same total value in the account at the end of the 60th month. [3]



<http://go.vjc.sg/2018p1q11>

Qn	Solution			Comments
11ia	n	Mth	Amount @ end of month	Read question carefully, the total in the account increases by 0.2%, not 0.2. For questions involving money, we usually round off to nearest cent unless otherwise stated.
	1	Jan 2016	$1.002(100)$	
	2	Feb 2016	$1.002^2(100)$	
	3	Mar 2016	$1.002^3(100)$	
		$\vdots$		
	12	Dec 2016	$1.002^{12}(100)$	
	Amount worth at 31 Dec 2016 $= \$1.002^{12}(100) = \102.43			
b	Method 1			Method 1 is what we are used to doing in the tutorials.
n	Mth	Beginning	End	
1	Jan 2016	100	$1.002(100)$	
2	Feb 2016	$1.002(100) + 100$	$1.002^2(100) + 1.002(100)$	
3	Mar 2016	$1.002^2(100) + 1.002(100) + 100$	$1.002^3(100) + 1.002^2(100) + 1.002(100)$	
	$\vdots$			
Total amount in account at 31 Dec 2016 (end of 2016) $= 100(1.002 + 1.002^2 + \dots + 1.002^{12})$ $= \frac{100(1.002)(1.002^{12} - 1)}{1.002 - 1}$ $= \$1215.71$				

Qn	Solution	Comments																
	Method 2 The first \$100 is worth $1.002^{12}(100)$. The second \$100 is worth $1.002^{11}(100)$. ... The last \$100 is worth $1.002(100)$. Total amount in account at 31 Dec 2016 $= 100(1.002 + 1.002^2 + \dots + 1.002^{12})$ $= \frac{100(1.002)(1.002^{12} - 1)}{1.002 - 1}$ $= \$1215.71$	Method 2 is making use of part (i)'s idea.																
c	Checking amount at the end of the month, $\frac{100(1.002)(1.002^n - 1)}{1.002 - 1} > 3000$ By GC, <table border="1"><tr><th>n</th><th>Amount worth at the end of the month</th></tr><tr><td>29</td><td>2988.65</td></tr><tr><td>30</td><td>3094.82</td></tr></table> The total amount in the account exceeds \$3000 in Jun 2018 Amount in account on 1st Jun 2018 = $2988.65 + 100$ $= 3088.65 > 3000$ The total amount in the account exceeds \$3000 at the 1st Jun 2018.	n	Amount worth at the end of the month	29	2988.65	30	3094.82	What you key into GC:  If the amount reaches \$3000 after adding 100, then it happened at the start. Otherwise, it happened at the end.										
n	Amount worth at the end of the month																	
29	2988.65																	
30	3094.82																	
ii	Amount worth at 31 Dec 2016 $= \$ (100 + 12b)$	Read question carefully, only the \$100 will generate bonus. The accumulated bonus does not earn bonus.																
b	Method 1 <table border="1"><tr><th>n</th><th>Mth</th><th>Beginning</th><th>End</th></tr><tr><td>1</td><td>Jan 2016</td><td>100</td><td>$100 + b$</td></tr><tr><td>2</td><td>Feb 2016</td><td>$200 + b$</td><td>$200 + b + 2b$</td></tr><tr><td>3</td><td>Mar 2016</td><td>$300 + b + 2b$</td><td>$300 + b + 2b + 3b$</td></tr></table> Total amount in account at 31 Dec 2017 $= 2400 + b + 2b + \dots + 24b$ $= 2400 + \frac{24}{2}(b + 24b)$ $= 2400 + 300b$	n	Mth	Beginning	End	1	Jan 2016	100	$100 + b$	2	Feb 2016	$200 + b$	$200 + b + 2b$	3	Mar 2016	$300 + b + 2b$	$300 + b + 2b + 3b$	Method 2 might be more intuitive.
n	Mth	Beginning	End															
1	Jan 2016	100	$100 + b$															
2	Feb 2016	$200 + b$	$200 + b + 2b$															
3	Mar 2016	$300 + b + 2b$	$300 + b + 2b + 3b$															

Qn	Solution	Comments
	Method 2 On 31 Dec 2017, Amount first \$100 is worth = $100 + 24b$ Amount second \$100 is worth = $100 + 23b$... Amount last \$100 is worth = $100 + b$ Total amount in account at 31 Dec 2017 $= 2400 + b + 2b + \dots + 24b$ $= 2400 + \frac{24}{2}(b + 24b)$ $= 2400 + 300b$ $2400 + 300b = 2800$ $b = 1.33$	
iii	$\frac{100(1.01)(1.01^{60} - 1)}{1.01 - 1} = 6000 + \frac{60}{2}(b + 60b)$ $1830b = 2248.6367$ $b = 1.23$	